

Shivang M Raval

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PROFESSIONAL SUMMARY

M.S. Computer Science candidate at Northeastern University specializing in AI/ML infrastructure, LLM systems, and multi-agent architectures. Google Cloud Certified Machine Learning Engineer with a published paper on ML deployment strategy and hands-on experience shipping production inference systems at scale.

EDUCATION

Northeastern University, Khoury College of Computer Sciences Boston, MA
M.S. Computer Science, GPA: 3.6/4.0 — Expected June 2027
Pandit Deendayal Energy University Gandhinagar, India
B.Tech. Computer Science, GPA: 3.5/4.0 — 2021–2025

EXPERIENCE

Research Participant, Multi-Agent Systems & Real-Time AI 2026
Google Developer Group (GDG) Cloud Boston — BuildWithAI Workshop

- Built production-grade multi-agent architectures on GCP using Google's Agent Development Kit (ADK) and the A2A protocol, with quality control via Vertex AI Evaluation Service and Model Armor.
- Engineered a bi-directional streaming agent using the Gemini Live API, Cloud Spanner Graph RAG, and Vertex AI Memory Bank.

AI Engineering Intern → AI Department Manager Jan 2025 – Jun 2025
Webeart AI

- Shipped ML inference systems serving 100K+ daily requests at sub-500ms latency; ran studies on model generalization and RLHF reward modeling.
- Promoted to lead a 20-engineer research team; diagnosed bottlenecks in large-scale ML pipelines and drove changes that raised research throughput by 50%.

Data Science Intern Jun 2024 – Jul 2024
Tor.ai

- Applied regression, ensemble methods, and cross-validation to a forecasting problem, achieving an MAE of 0.12 — 20% better than baseline.

PROJECTS

Distributed ML Training Platform

- Built distributed training infrastructure achieving a 10.6× speedup (8h → 45min) via PyTorch DDP; containerized with Docker/Kubernetes and automated CI/CD, cutting model refresh from 4 days to 3 hours across 8 workflows.

Production RAG Infrastructure

- Built a low-latency RAG pipeline on Vertex AI and Gemini with a FastAPI serving layer, Redis caching (60% hit rate, sub-10ms cached latency), and FAISS vector indexing.

LLM Inference Engine

- Implemented continuous batching, paged KV-cache, INT8 quantization, and speculative decoding in a custom serving engine; benchmarked a genuine 1.64× latency reduction from continuous batching alone under concurrent load.

AI Infrastructure Stack: Entity Resolution, RL Gym & Policy Classifier

- Fine-tuned a multilingual encoder (InfoNCE loss) for cross-lingual entity resolution across three languages; reused the encoder as a reward oracle for a PPO agent and as the backbone for a zero-shot Gemini-based policy classifier.

CERTIFICATIONS

- Google Cloud Certified: Machine Learning Engineer

SKILLS

Languages: Python, C++, Java, TypeScript, OCaml, SQL, Bash
ML & Data: PyTorch, RLHF, FAISS, LangChain, LangGraph, Redis, PostgreSQL
Cloud & Infrastructure: GCP (Vertex AI, ADK, Gemini API), AWS, Docker, Kubernetes, SageMaker, CI/CD
Systems & Agent Protocols: Distributed Systems, Concurrency, Model Context Protocol (MCP), A2A Protocol

PUBLICATIONS & LEADERSHIP

- IJISRT 2024: "ML Deployment Platforms in Business: Scalable Infrastructure, Operational Challenges, and Deployment Strategies"
- GDG Cloud Boston — Member, BuildWithAI 2026 Participant